

Mahmoud Bendari

647-879-7640 | mbendari@uoguelph.ca | [in](#) LinkedIn | [G](#)itHub | mahmoudbendari.com

EDUCATION

University of Guelph

Bachelor of Engineering, Computer Engineering Co-op

Sept. 2025 – Apr. 2030

Guelph, ON

TECHNICAL SKILLS

Languages & Frameworks: C/C++, Java, Python, JavaScript, Node.js, React.js, Next.js, HTML/CSS

Tools & Protocols: PostgreSQL, Redis, Linux, PlatformIO, Git, REST APIs, I2C, SPI, UART, PWM

Hardware & CAD: AutoCAD, SolidWorks, KiCad, Ultimaker Cura, Soldering, Adobe Creative Cloud

Soft Skills: Time Management, Client Communication, Cross-Functional Teamwork, Problem Solving, Arabic

EXPERIENCE

Software Engineering Intern

Pearl Travel

June 2025 – Aug. 2025

Alexandria, Egypt (Remote)

- Designed and developed a full-stack B2C booking platform using React.js and Node.js to modernize the agency's manual workflows, successfully onboarding over 5,000 new user accounts.
- Built backend REST APIs with Express.js and integrated a PostgreSQL database with Redis caching, which sped up flight search queries by 40% and significantly reduced manual processing time for travel agents.
- Collaborated with management using Agile methodologies to refine the UI/UX checkout flow, improving overall user retention and directly contributing to \$20,000 in new online revenue.

Start-up Founder

Digital Fixers

Jan. 2025 – Sept. 2025

Guelph, ON

- Deployed custom web solutions and local marketing integrations using HTML/CSS/JS and supporting libraries directly interfacing with B2B/B2C clients to capture operational requirements and increasing visibility by 90%.
- Engineered a rigorous hardware diagnostic workflow utilizing multimeters and debugging tools to systematically troubleshoot complex electromechanical and board-level system failures.
- Optimized backend inventory logistics by developing Excel models to track real-time expenses and manage global supplier orders, maintaining 100% reliable component availability.

PROJECTS

Dual-Axis Solar Tracker | Embedded & Firmware Developer

June 2026 – Aug. 2026

- Designed C++ firmware for an Arduino Nano ESP32 to dynamically control dual servo motors via PWM, processing analog light sensor inputs to maximize energy capture efficiency by up to 30%.
- Engineered custom PCB layouts and circuitry schematics in KiCad, and modelled a precise 3D-printed electromechanical chassis in SolidWorks and Ultimaker Cura, optimizing spatial layout for LDRs and an I2C-driven OLED telemetry display.
- Debugged complex hardware power routing and isolated electrical noise to ensure 100% reliable continuous sensor polling under varying environmental light conditions.

EcoLens (GenAI Genesis Hackathon) | Software Engineer

Mar. 2026

- Architected EcoLens full-stack application alongside a 3-person engineering team, enabling users to scan grocery items and dynamically calculate product-specific environmental externality costs.
- Engineered a rapid Python data pipeline utilizing REST APIs to parse complex product component datasets, calculating real-time sustainability metrics and streaming JSON responses to drive eco-conscious consumer decisions.
- Deployed functional MVP within a 48-hour time constraint utilizing Agile methodologies, managing version control via Git and finalizing a successful project demonstration on Devpost.

University IT Helpdesk System | Software Engineer (Java)

Feb. 2026 – Apr. 2026

- Architected an object-oriented IT helpdesk ticketing system in Java alongside a collaborative engineering team, utilizing inheritance and polymorphism to efficiently categorize and route support requests.
- Managed version control and project workflows utilizing Git and GitHub, orchestrating branch merges and conducting code reviews to ensure seamless component integration across multiple programmers.
- Engineered scalable class hierarchies to manage varying user permission levels and ticket resolution statuses, demonstrating strong enterprise software design principles and strict encapsulation.